

visualization Tool for Electric Vehicle Charge and Range Analysis
using Tableau

TEAM ID	SWTID-2026-6767
PROJECT TITLE	visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	2 MARKS

5.2 – Sprint Delivery Plan

Sprint 1: Project Setup & Requirement Analysis

Duration: Week 1

Tasks:

- Define project objectives and scope.
- Identify EV charge and range analysis requirements.
- Collect the ***cheapest electriccars-evdatabase*** dataset.
- Install XAMPP, MySQL Workbench, and Tableau.

Deliverables:

- Project requirements document.
- Software environment setup completed.

Sprint 2: Database Creation & Data Import

Duration: Week 2

Tasks:

- Create **electric_vehicle** schema in MySQL.
- Import dataset using **Table Data Import Wizard**.
- Verify data integrity and structure.
- Create necessary database tables.

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Deliverables:

- EV dataset successfully stored in MySQL database.
- Database schema ready for analysis.
- **Sprint 3: Data Cleaning & Preparation**

Duration: Week 3

Tasks:

- Remove duplicate records.
- Handle missing and null values.
- Validate data accuracy.
- Standardize column formats and data types.

Deliverables:

- Clean and validated EV dataset.
- Data quality report.

Sprint 4: Tableau Connectivity & Data Extraction

Duration: Week 4

Tasks:

- Install and configure MySQL Connector.
- Connect Tableau to MySQL database.
- Create Tableau Extract.
- Verify data synchronization.

Deliverables:

- Successful Tableau-MySQL connection.
- Optimized Tableau Extract file.

Sprint 5: EV Data Analysis

Duration: Week 5

Tasks:

- Analyse battery capacity data.
- Analyse driving range performance.

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- Evaluate energy consumption efficiency.
- Identify trends and patterns.

Deliverables:

- Analysis results and KPI definitions.
- Insights report.

Sprint 6: Dashboard Development

Duration: Week 6

Tasks:

- Design dashboard layout.
- Create charts, graphs, and KPIs.
- Add filters and interactive features.
- Develop EV performance visualizations.

Deliverables:

- Interactive Tableau Dashboard.
- Range and charge analysis reports.

Sprint 7: Testing & Validation

Duration: Week 7

Tasks:

- Perform Functional Testing.
- Conduct Performance Testing.
- Validate dashboard calculations.
- Fix identified issues.

Deliverables:

- Test report.
- Verified dashboard outputs.

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Sprint 8: Documentation & Final Delivery

Duration: Week 8

Tasks:

- Prepare project documentation.
- Create presentation slides.
- Document results, conclusions, and future scope.
- Final project review and deployment.

Deliverables:

- Final project report.
- Project presentation.
- Tableau dashboard ready for demonstration.

Sprint Summary

Sprint	Activity	Duration
Sprint 1	Project Setup & Requirement Analysis	Week 1
Sprint 2	Database Creation & Data Import	Week 2
Sprint 3	Data Cleaning & Preparation	Week 3
Sprint 4	Tableau Connectivity & Data Extraction	Week 4
Sprint 5	EV Data Analysis	Week 5
Sprint 6	Dashboard Development	Week 6
Sprint 7	Testing & Validation	Week 7
Sprint 8	Documentation & Final Delivery	Week 8

This sprint plan follows the actual workflow of your project using **XAMPP**, **MySQL Workbench**, **MySQL Connector**, and **Tableau** with the **electric_vehicle** database and **cheapest electriccars-evdatabase** dataset.